SATS

PHASE-BOUNDED GAIT EVENT DETECTION AND PREDICTION DURING STW TRANSITION USING MACHINE LEARNING

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Abstract:

This project presents the design and implementation of a wearable-only gait phase classification and prediction framework integrating IMU-based event detection with phase-bounded EMG feature extraction and machine learning models. A complete signal processing pipeline was developed to ensure reliable and physiologically meaningful feature generation.

Random Forest was used for gait phase classification, while LSTM networks were applied for next-event prediction using subject-wise cross-subject evaluation. The system achieved high accuracy under real-world wearable conditions, demonstrating the feasibility of robust, sensor-based gait phase recognition for rehabilitation and assistive applications.

Problem Statement:

Wearable-based gait phase detection systems operating outside controlled laboratory environments face significant challenges due to sensor noise, motion artifacts, inter-subject variability, and temporal ambiguity between gait phases. Disturbances such as electrode movement, powerline interference, and heterogeneous sampling rates degrade EMG signal quality and complicate reliable phase segmentation.

The problem addressed in this project is achieving accurate and subject-independent gait phase classification and next-event prediction using wearable EMG and IMU signals through robust signal processing and machine learning–based modeling.

Methodology:

- **System Design**:

Wearable **EMG (8 channels)** + foot & Trunk **IMU,** foot IMU for event detection, EMG for muscle feature extraction.

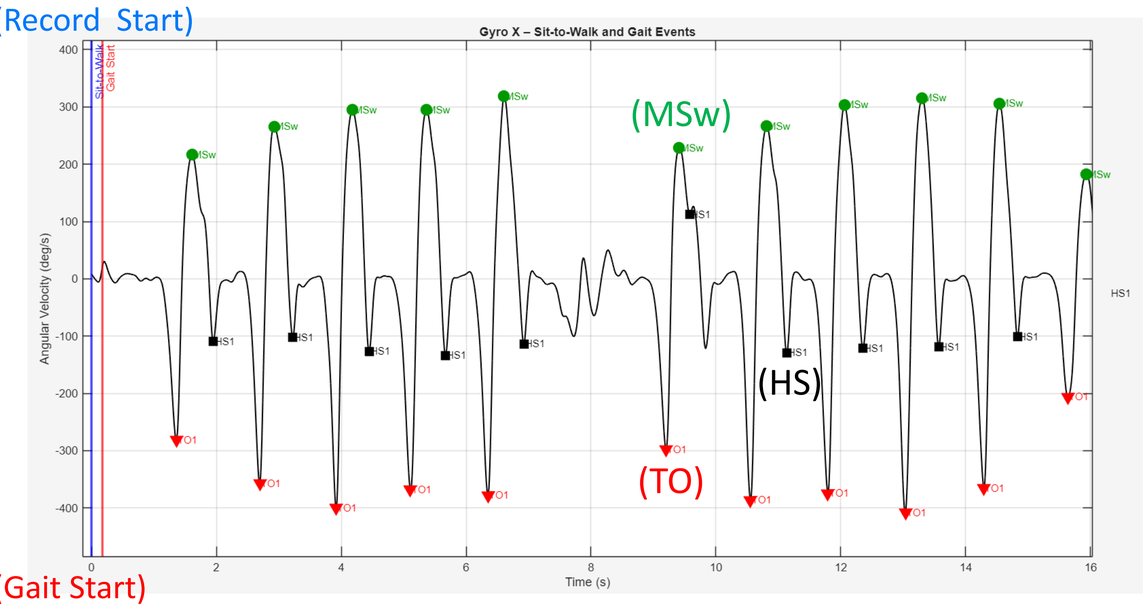
- **Signal Processing:**

For IMU: NaN/Inf removal, low pass **6 Hz FC**, Rectify.

For EMG: NaN/Inf removal, Resampling 1259 to 1778, Band-pass: **20–450 Hz, 50 Hz** notch filtering, Rectify, MVC normalization.

Phase-bounded segmentation (IMU-driven).

(Record Start)



(Gait Start)

- **Software Framework:**

Modular MATLAB pipeline, Automated feature extraction, Structured subject-wise dataset, Reproducible ML workflow.

- **Features & Models:**

Features (64 total):

RMS, MAV, IEMG, WL, ZC, SSC, PF, TP for **8 Muscles**.

Models:

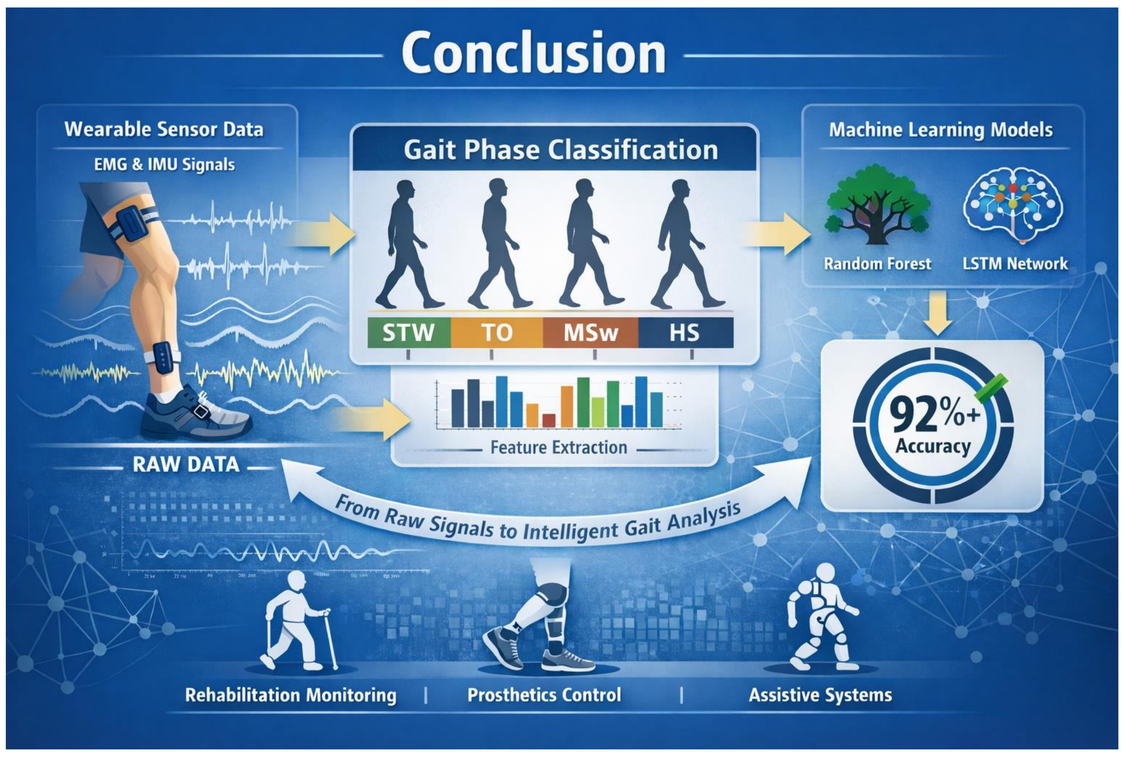
Random Forest → Event classification.

LSTM → Next-event prediction.

- **Experimental Validation:**

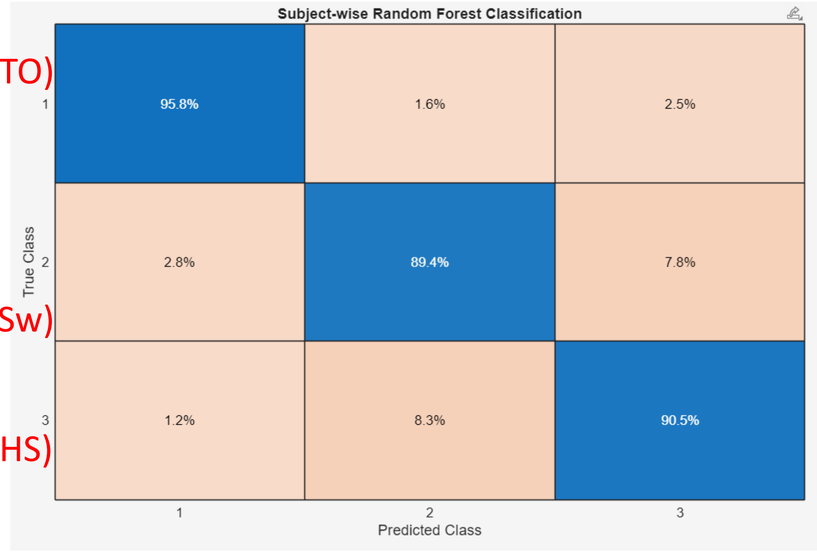
65 subjects → **55 train / 10 test**.

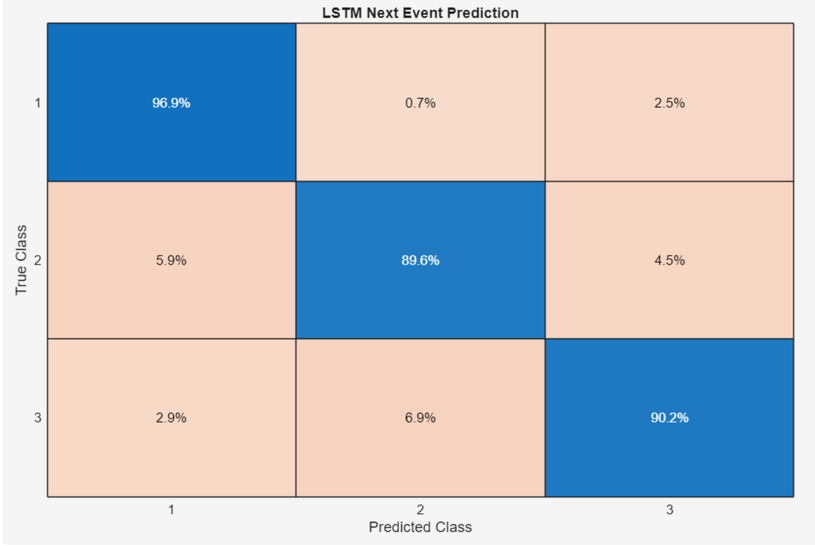
Evaluated using: Accuracy, Precision, Recall, F1-score, Confusion Matrix, Multi-class evaluation (TO, MSw, HS).



Results and Conclusion:

- High classification performance achieved (RF & LSTM).
- RF = **91.91%**, LSTM = **92.03%**.
- Strong subject-independent generalization.
- Phase-bounded segmentation enhanced model accuracy.
- Temporal dynamics successfully modeled using LSTM.





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